

Team members:

Part 1 – Choosing the right model

For each scenario below:

1. Identify the response variable.
2. Identify the variable type for the response variable.
3. Identify whether linear or logistic regression would be an appropriate model.
4. Identify whether RMSE or Accuracy should be used to assess predictive performance of the model.

Question 1

Predicting the number of hours a student studies per week using: major, number of classes, GPA.

Response variable:

Response variable type:

Regression model:

Evaluation metric:

Question 2

Predicting whether a customer will cancel their subscription using: monthly cost and subscription type.

Response variable:

Response variable type:

Regression model:

Evaluation metric:

Question 3

Using humidity and wind speed to predict daily temperature.

Response variable:

Response variable type:

Regression model:

Evaluation metric:

Question 4

Using attendance, major, and hours studied to predict if a student will pass or fail a course.

Response variable:

Response variable type:

Regression model:

Evaluation metric:

Question 6

Fill in the blanks:

For continuous response variables (or at least discrete numeric response variables that can take many values), _____ regression is appropriate and we can compute _____ to assess predictive performance of our model. A **larger** / **smaller** value of this is better. (circle one)

For binary response variables, _____ regression is appropriate and we can compute _____ to assess predictive performance of our model. A **larger** / **smaller** value of this is better. (circle one)

Understanding predictions**Question 7**

A linear regression model predicts (circle one): **value of y_i** / **log odds of y_i**

A logistic regression model predicts (circle one): **value of y_i** / **log odds of y_i**

We transform the output of the logistic regression model predictions to get predicted probability of $Y_i = 1$.

Question 8

To convert predicted probabilities into a final prediction, we use a cutoff (e.g., 0.5).

Using a cutoff of 0.4, what would the predicted value of y_i be for each predicted probability:

- $\hat{\pi}_i = 0.80 \rightarrow$ predicted value of $y_i =$
- $\hat{\pi}_i = 0.25 \rightarrow$ predicted value of $y_i =$
- $\hat{\pi}_i = 0.5 \rightarrow$ predicted value of $y_i =$